VIRTUAL MACHINE







INTRODUCTION

- In computing, a virtual machine (VM) is an emulation of a computer system.
- It runs on your current operating system (the host operating system) and provides virtual hardware to guest operating systems.
- From the guest operating system's perspective, the virtual machine is a real, physical computer.
- The virtual machine's emulation engine, called a hypervisor, handles the virtual hardware, including a CPU, memory, hard drive, network interface, and other devices.

- The virtual hardware devices provided by the hypervisor map to real hardware on your physical machine.
- For example, a virtual machine's virtual hard disk is stored in a file located on your hard drive.

VM TYPES

- **System virtual machines (also termed full virtualization VMs) provide a substitute for a real machine. They provide functionality needed to execute entire operating systems.**
- For example virtualbox, vmware player, Qemu, etc.

- **Process virtual machines** sometimes called an application virtual machine, runs as a normal application inside a host OS and supports a single process. It is created when that process is started and destroyed when it exits.
- *For example Wine software in Linux helps to run Windows application.*

POPULAR VM

- Virtualbox - It is opensource and cross platform software for virtualization. You can install it on window, linux, mac and BSD OS.
- Vmware- It is similar to virtualbox with some added features. But it is not free.
- Qemu- In addition to providing normal VM support, it is designed to also emulate different architectures and allow execution of software applications and operating systems written for another CPU or architecture.

THANKS